Customs Tax Calculator

By KRA

https://sbx.kra.go.ke/api/v1/taxCalcu…

Simulate API

Documentation

Tax Calculator API — Developer Reference

Base URL: `https://sbx.kra.go.ke`

Version: v1

Auth: Bearer token (see Authentication)

Endpoint Summary

#	Method	Path	Purpose
1	`GET`	`/v1/token/generate?grant_type=client_credentials`	Obtain access token
2	`POST`	`/api/v1/taxCalculator/searchByHSCode`	Search commodities by HS code
3	`POST`	`/api/v1/taxCalculator/searchByDesc`	Search commodities by description
4	`GET`	`/api/v1/taxCalculator/details`	Get commodity details and tax rates
5	`POST`	`/api/v1/taxCalculator/calculate`	Calculate customs tax

Authentication

All endpoints except token generation require a Bearer token in the `Authorization` header:

```
Authorization: Bearer {access_token}
```

Endpoints

1. Generate Access Token

Authenticates the caller and returns a time-limited access token required by all other endpoints.

```
GET https://sbx.kra.go.ke/v1/token/generate?grant_type=client_credentials
```

Refer to Authorization API Documentation

2. Search Commodity Codes by HS Code

Returns a list of commodities matching a given HS code prefix. The code sample must be **at least 4 digits**.

```
POST /api/v1/taxCalculator/searchByHSCode
```

Request Headers

Header	Value	Required
`Authorization`	`Bearer {access_token}`	Yes
`Content-Type`	`application/json`	Yes

Request Body

```
{
  "HSCode": "0101"
}
```

Field	Type	Required	Description
`HSCode`	string	Yes	HS code prefix for searching. Minimum 4 digits.

Response Body

Field	Type	Mandatory	Description
`success`	integer	Yes	Search status: `1` = results found, `0` = no results
`hscodesList`	array	Yes	Array of matching commodity objects (present when `success = 1`)
`hscodesList[].nmctcode`	string	Yes	Commodity code
`hscodesList[].nmctdesc`	string	Yes	Human-readable commodity description
`hscodesList[].stmucode`	string	Yes	Unit of measurement for the commodity
`hscodesList[].taxDetails`	array	Yes	List of applicable tax rates for this commodity

Example Response

```
{
  "success": 1,
  "hscodesList": [
    {
      "nmctcode": "01011010",
      "nmctdesc": "Live horses - Pure-bred breeding animals",
      "stmucode": "KG",
      "taxDetails": [ ... ]
    }
  ]
}
```

Response Codes

Code	Status	Description
`200`	OK	Search executed successfully
`400`	Bad Request	`HSCode` is fewer than 4 digits
`401`	Unauthorized	`access_token` is missing or invalid

3. Search Commodity Codes by Description

Returns a list of commodities whose description matches the given search text.

```
POST /api/v1/taxCalculator/searchByDesc
```

Request Headers

Header	Value	Required
`Authorization`	`Bearer {access_token}`	Yes
`Content-Type`	`application/json`	Yes

Request Body

```
{
  "desc": "live horses"
}
```

Field	Type	Required	Description
`desc`	string	Yes	Keyword or phrase to match against commodity descriptions

Response Body

Same schema as Search by HS Code — returns a `success` flag and a `hscodesList` array.

Response Codes

Code	Status	Description
`200`	OK	Search executed successfully
`401`	Unauthorized	`access_token` is missing or invalid

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4. Get Commodity Details and Tax Rates

Retrieves full details and all applicable tax rates for a specific commodity using its exact code.

```
GET /api/v1/taxCalculator/details
```

Request Headers

Header	Value	Required
`Authorization`	`Bearer {access_token}`	Yes
`Content-Type`	`application/json`	Yes

Request Body

```
{
  "nmctcode": "01011010"
}
```

Field	Type	Required	Description
`nmctcode`	string	Yes	Exact commodity code to retrieve full details for

Response Body

Field	Type	Mandatory	Description
`details.nmctcode`	string	Yes	Commodity code
`details.nmctdesc`	string	Yes	Commodity description
`details.stmucode`	string	Yes	Unit of measurement
`details.taxDetails`	array	Yes	Array of tax rate objects
`details.taxDetails[].taxType`	string	Yes	Category of tax (e.g., Import Duty, VAT)
`details.taxDetails[].taxDesc`	string	Yes	Human-readable tax rate description
`details.taxDetails[].taxRate`	number	Yes	Numeric tax rate value
`details.taxDetails[].calcMethod`	string	Yes	Calculation method: `per unit` or `percentage`
`details.taxDetails[].calcCurrency`	string	Yes	Currency code for the tax rate
`details.taxDetails[].movementType`	boolean	Yes	`true` = export, `false` = import

Example Response

```
{
  "details": {
    "nmctcode": "01011010",
    "nmctdesc": "Live horses - Pure-bred breeding animals",
    "stmucode": "KG",
```

```
    "taxDetails": [
      {
        "taxType": "IDF",
        "taxDesc": "Import Declaration Fee",
        "taxRate": 3.5,
        "calcMethod": "percentage",
        "calcCurrency": "KES",
        "movementType": false
      }
    ]
  }
}
```

Response Codes

Code	Status	Description
`200`	OK	Commodity details retrieved successfully
`401`	Unauthorized	`access_token` is missing or invalid

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5. Calculate Customs Tax

Calculates the full customs duty and levy breakdown for a commodity. All response values are expressed in **Kenyan Shillings (KES)**.

```
POST /api/v1/taxCalculator/calculate
```

Request Headers

Header	Value	Required
`Authorization`	`Bearer {access_token}`	Yes
`Content-Type`	`application/json`	Yes

Request Body

Field	Type	Required	Description
`value`	number	Yes	Commodity value expressed in the specified currency
`quantity`	number	Yes	Quantity based on the commodity's unit of measurement
`currency`	string	Yes	ISO 4217 currency code for `value` (e.g., `USD`, `EUR`)
`movement`	string	Yes	Trade direction. Must be **uppercase**: `IM` (import) or `EX` (export)

Field	Type	Required	Description
`nmctcode`	string	Yes	Commodity code to be calculated
`complementary`	object	No	Additional data required by certain commodity types

Example Request

```
{
  "value": 5000,
  "quantity": 10,
  "currency": "USD",
  "movement": "IM",
  "nmctcode": "01011010",
  "complementary": {}
}
```

Response Body

Field	Type	Mandatory	Description
`CustomsValueKes`	number	Yes	Customs value of the commodity converted to KES
`CurrencyValue`	number	Yes	Exchange rate of the given currency against KES
`Tax`	number	Yes	Total customs tax payable in KES
`Levies`	number	No	Combined IDF and RDL levy total in KES
`IDF`	number	No	Import Declaration Fee in KES
`RDL`	number	No	Railway Development Levy in KES

Example Response

```
{
  "CustomsValueKes": 650000.00,
  "CurrencyValue": 130.00,
  "Tax": 97500.00,
  "Levies": 26000.00,
  "IDF": 22750.00,
  "RDL": 3250.00
}
```

Response Codes

Code	Status	Description
`200`	OK	Tax calculated successfully

Code	Status	Description
`400`	Bad Request	`movement` is not uppercase `IM` or `EX`
`401`	Unauthorized	`access_token` is missing or invalid

Error Handling

Code	Cause	Resolution
`400`	Invalid parameter format	Check that all parameters meet format constraints — `movement` must be `IM` or `EX` in uppercase; `HSCode` must be at least 4 digits.
`401`	Unauthenticated request	Ensure a valid, non-expired Bearer token is included in the `Authorization` header. Re-authenticate if necessary.

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